INCH-POUND

MIL-R-43838B(GL)
25 May 1993
SUPERSENDING.
MIL-R-4383A(GL)
31 March 1983

MILITARY SPECIFICATION

RELEASE, CARGO PARACHUTE, AIRDROP, M-2

This specification is approved for use by the Natick Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of cargo parachute release assembly consisting of a release mechanism, upper and lower suspension link assemblies and cargo parachute connectors. The cargo parachute connectors may be procured sepately (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specificaitons, standards, and handbooks. The followings specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are thoseslisted in the issue of the Department of Defense Index or Specifications and Standards (DODISS) and supplement therto, sited in the solicitation (see 6.2)

SPECIFICATIONS

MILITARY

MIL-H-6875 - Heat Treatment of Steel Process For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document hould be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5017, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1670

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-207 - DoD Materiel Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Goverment documents, drawings, and publicaitons. The fallowing other Goverment documents, drawings, and publicaitons form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

11-1-511 - Toggle
11-1-513 - Plate, Front
11-1-565 - Release Assembly, Cargo Parachute, Airdrop. M-2
11-1-566 - Plate, Back
11-1-894 - Timer Delay Assembly (Release Cargo Parachute, M1 & M2)
11-1-1487 - Release Assembly, Cargo Parachute, Airdrop, M-1

(Copies of drawings are available from the U.S. Army Natick Research, Development and Engineering center, ATTN: SATNC-UXT, Natick, MA 01760-5017).

2.2 Non-Government publicaitons. The following documents from a part of this document to the exxtent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted ar those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

b46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

(Application for copies should be addressed to the American National Standards Institute, Inc., 11 West 42nd Street, New York, NY 10036.)

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(Non-Govenment standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informtional services.)

2.3 Order of precedence. In the event of a conflict between teh text of this document and the references cited herein, the text of this documents takes precedence. Nothing in this document. however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Materials and components. The materials and components used in the manufacture of the M-2 cargo parachute release shall conform in all respects to the requirements specified or shown on Drawing 11-3-565, and all subsidiary drawings and parts lists pertaining thereto, and as specified herein.

3.3 Construction and supplementary process. The construction and supplementary process used in the manufacture of the cargo parachute releases shall conform to the requirements specified or shown on Drawing 11-1-565, and all subsidiary drawings and parts lists pertaining thereto, and as specified herein.

3.3.1 Surface finish. American National Standard Institute (ANSI) standard No. 346.1, shall apply for the interpretation of surface roughness values specified on the applicable drawings.

3.4 Performance.

3.4.1 Cargo parachute connector proofload. The cargo parachute connector shall show no visual evidence of failure (fracture or permanent deformation) when subjected to a tensile proofload of not less than 25,000 pounds for not less than 15 seconds when tested as specified in 4.4.5.

3.4.2 Release assembly proofload. The release assembly shall show no visual evidence of failure (fracture or permanent deformatin) when subjected to a tensile proofload of not less than 92,000 pounds for not less than 15 seconds when tested as specified in 4.4.5.

3.4.3 Release assembly unlocking. At completion of the timer unwinding cycle, the timer delay with attached toggle locking slides shall drop to the unlocked bottom position with no evidence of interference when tested as specified in 4.4.5.

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3.5 Identificaiton marking. Identification marking of release assemblies and all components procured separately shall be in accordance with MIL-STD-130 and with the legend and location as shown on applicable drawings,

3.6 Goverment furnished property(GFP). The following items, when required for proofload testing of the release assembly or separately procured cargo parachute connectors, may be oftained from the procuring activity on a loan basis (see 6.4).

Release Assembly Cargo Parachute, Airdrop M-1 or M-2
Part No. 11-1 -1487-1 or 11-1-1487-2
11-1-565-1 or 11-1-565-2

Timer Delay Assembly; Releases, Cargo, Parachute, M-1 and M-2
Part No. 11-1-894-1

3.7 Workmanship. The release assembly and its components shall conform to the quality of product established by this specification and the occurance of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspeciton requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase oder, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved the Goverment. The Goverment reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall met all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all procucts or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is all acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be exmined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to insure compliance with all dimensional requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follcws:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and shall be tested for the characteristics specified in 4.4.5. Any nonconformance shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amentd, modified, or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

4.4.2.1 In-process heat treatment inspection. Heat treatd components shall be inspected to determine compliance with requirements specified on the applicable drawings. The provisions of MIL-H-6875 applicable to routine quality control shall apply, except that the inspection level shall be S-2 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

4.4.2.2 In-process simensional examination. Prior to assembly, but after heat treatment and application of protective finish; the items listed below shall be examined for conformance to the dimensions indicated. The lot size shall be expressed in units of items of applicable type. The sample unit shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

Item	Drawing	Dimensions
Toggle	11-1511	.313 and .520
Front plate	11-1-513	.317 and .520
Back plate	11-1-566	.317 and .520

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table 1. The lot size shall be expressed in units of complete release assemblies or cargo parachute connectors, as applicable. The sample unit shall be one complete release assembly or eight cargo

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parachute connectors, asapplicable. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for total (major and minor ccmbined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Omitted or not type specified	101	
	Coating powdery, coarse grained, mottled, not adherent, evidence of corrosion	102	
	Supplementary treatment omitted, not type Specified, tacky	103	
	Color not as specified		201
	Dirt, oil, grease or other foreign matter		202
Design	Any departure from design specified on drawings	104	
Construstion and workmanship	Surface roughness value in excess of rating specified on drawing	105	
	Any component missing, cracked, visually malformed, misaligned, damaged	106	
	Sharp edges, burrs or metal slivers	107	
	Components incorrectly assembled	108	
	Components loose	109	
	Movement of parts inhibited	110	
	Threaded sections damaged	111	
Identification	Missing, incomplete, illegible, improper location, or not as specified		203

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Anyu dimension not with the spe3cified tolerance shall be classified as a defect. The lot size shall be expressed in units of complete release assemblies or cargo parachute connectors, as applicable. The sample unit shall be one complete release assembly or eight cargo parachute connectors, as applicable. The inspection level shall S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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4.4.5 End item testing. Cargo parachute connectors and release assemblies (proofload and unlocking) shall be tested as specified in 4.5.1 and 4.5.2 for compliance with requirements of 3.4.1, 3.4.2 and 3.4.3 respectively. The lot size shall be no more than 50 completely assembled cargo parachute connectors or complete release assemblies. Th sample unit shall be one completely assembled cargo parachute connector or complete release assembly. Cargo parachute connectors showing any visual evidence of failure (fracture or permanent deformation) shall be cause for rejection of the lot. Release assemblies showing any visual evidence of failure (fracture or permanent deformation) of any component of the release assembly, exclusive of the cargo parachute connectors and time delay assembly, or failure of the delay assembly and attached lock slides to drop freely without interference to unlocked bottom position shall be cause for rejection of the lot.

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed helm. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping-container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
content	Number per container is more or less than reuired

4•4 l 7 Palletization examination. The fully packaged and palletized end items shall be examined for th defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Infect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified

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Weight	Exceeds load limits
Marking	Cmitted; incorrect; illegible; of improper size, location, sequence, or method of applicaiton

4.5 Methods of inspection

4.5.1 Cargo parachute connector proofload testing. The cargo parachute connector to be tested shall be installed in a contractor furnished test fixture. The test fixture shall be attached to a direct reading and calibrated tensile testing device capable of imposing a 25,000 pound tensile force on the test item. The test fixture shall be constructed essentially in accordance with figure 1 and designed with a minimum safety factor of 2.0. The connector sleeve may be removed to facilitate attachment of the connector body to the test fixture. Subject to Government approval, a mechanism may be developed to restrain the connector arms in the same manner in which they are normally attached to the release assembly in lieu of the release and time delay assemblies shown on figure 1. A proofload of 25,000 pounds shall be applied to the connector at a rate of not less than 3,000 pounds/second nor greater than 6,000 pounds/second. The connector shall withstand the tensile load for 15 seconds after which the load shall be reduced to zero. The proof load of 25,000 poounds shall be immediately re-applied, held for 15 seconds, and again reduced to zero.

4.5.2 Release assembly proofload and unlocking testing. The release assembly shall be assembled with eight cargo parachute connectors and with a timer delay assembly (GFP, see 6.4), installed in the locked position. The locked position requires raising the timer delay assembly with the attached toggle lock slides, turning the timer stem approximately 90 degrees clockwise to engage the timer keys with the release cavity slots, and inserting the arming wire through the slot provided and through the timer stem. The complete assembly shall then be mounted in a contractor furnished test fixture. The assembly shall then be mounted in a contractor furnished test fixture. The test fixture shall be attached to a calibrated tensile testing machine capable of imposing a 92,000 pound tensile force on the test fixture and release assembly. The test fixture shall be fabricated in accordance with figure 2 and designed with a minimum safety factor of 2.0. The fixture shall provide that the total applied tensile force be equally distributed among the eight cargo parachute connectors and also divided equally between the two lower ling assemblies. The cargo parachute connector sleeves may be removed to facilitate attachment of the connector bodies to the test fixture. The lower link assmblies shall be connected so that only one branch of each link assembly will be subjected to the tensile load. The proofload of 92,000 pounds shall be applied to the release assembly at a rate of not less than 5,000 pounds/second, nor greater than 10,000 pounds/second. The release assembly shall withstand the tensile load for 15 seconds, after which the tensile force shall be reduced to zero. The proofload of 92,000 pounds shall be immediately reapplied, held for 15 seconds, and then reduced to zero. After completion of the second proofload application, a tensile

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load not to exceed 100 pounds shall be applied to the release assembly. The release assembly shall then be activated by removing the arming wire. After the timing cycle is completed the delay assembly with attached toggle lock slides shall drop freely within the release body to the unlocked (bottom) position. The mile had shall then be reduced to zero and the release assembly disassembled for inspection.

4.5.3 Procurement of Cargo parachute connectors as a replacement Unit. The cargo parachute connectors when procured as a replacement or spare part (see 6.2) shall be tested for the requirements of 3.4.1 and conformance to 4.4.5. A complete and operable release assembly and timer delay assembly will be provided (see 3.6) for conduct of the proofload test, if required

5. PACKAGING

5.1 Preservation. The level of preservation shall be as required by the procuring activity in accordance with MIL-STD-2073-1.

5.2 Packing. 5.2 Packing. The level of packing shall be as required by the procuring activity in accordance with MIL-STD--2073-l.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Release, Cargo Parachute, Airdrop M-2, covered by this document is used to automatically disengage the parachute from the load at the moment of ground impact. This prevents the parachute from dragging or overturning the load. This release is intended for loads having a gross rigged weight of 15,000 pounds to 42,000 pounds.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3 and 6.3).
d. Levels of preservation and packing (see 5.1 and 5.2).
e. When a separate procurement of cargo parachute connectors is desired (see 1.1 and 4.5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-3 or 52.209-4. The first article should be a predrouction sample. The contracting offficer should specify the appropriate type of first article and the number of units to be furnished.

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The Contacting officer should also include specifiC instruction in acquisition documents regarding arrangments for selection, inspection, and approval of the first article.

6.4 Sample. For access to goverment furnished property (see 3.6) address the contracting activity issuing the invitation for bids or request for Proposal.

6.5 Subject term (key word listing)

Parachute mechanism

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify change with respect to the previous issue due to the extensiveness of the Changes.

Custodians:

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Review activity:

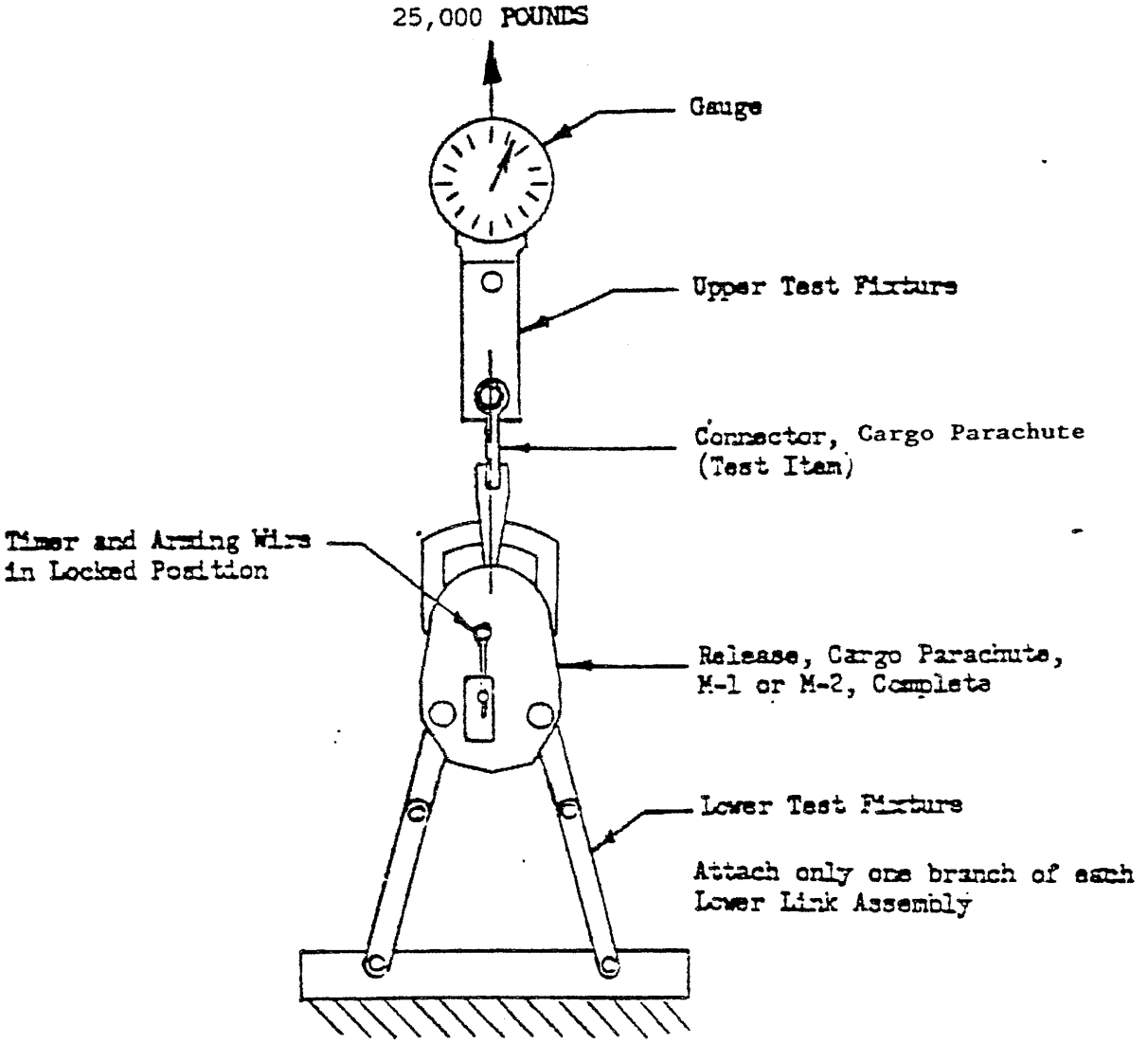
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Preparing activity:

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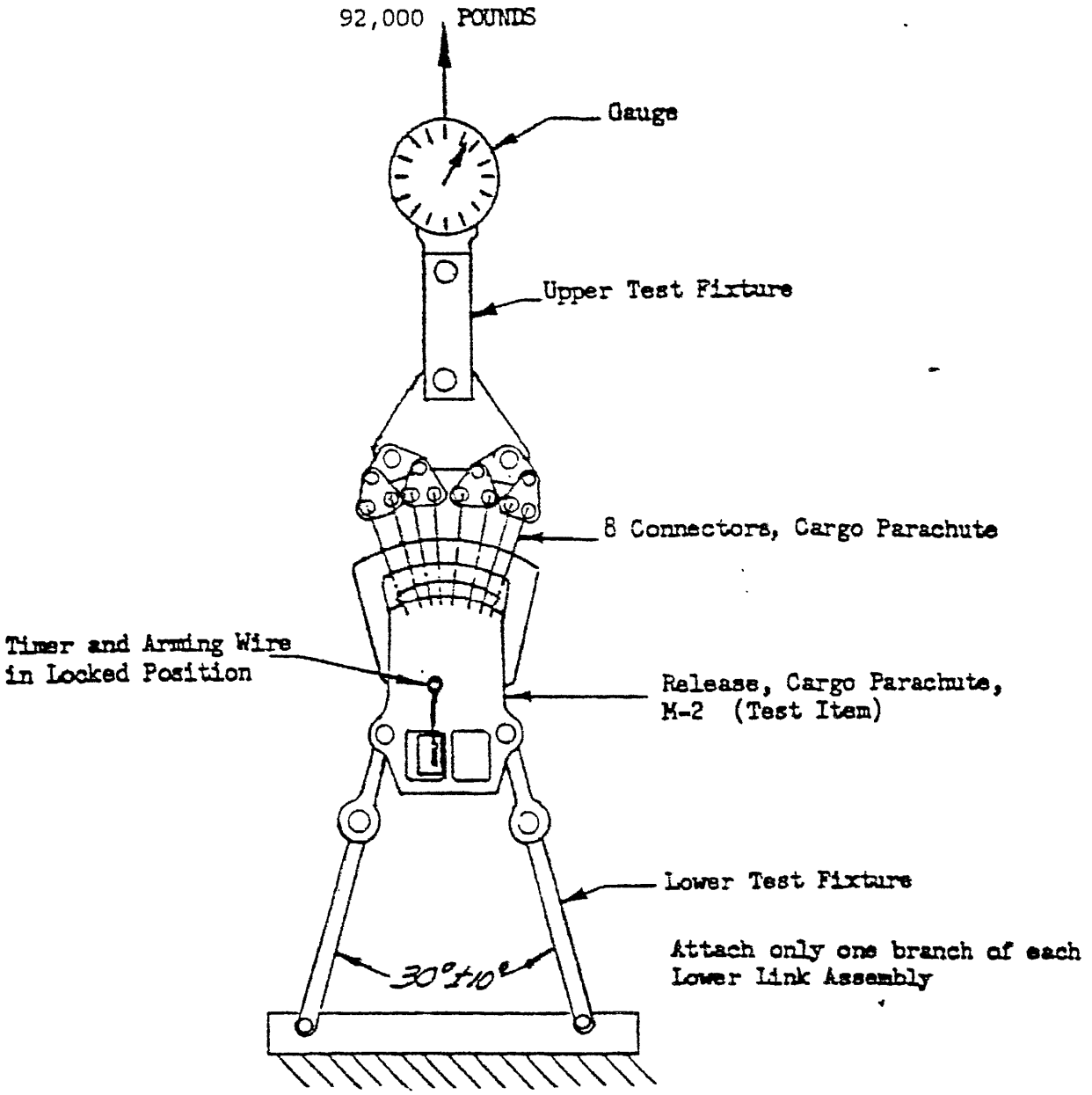
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PROOF LOAD CONNECTOR, CARGO PARACHUTE

FIGURE 1

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PROOF LOAD TEST, M-2

FIGURE 2

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-43838B(GL)	1993 May 25

3. DOCUMENT TITLE
RELEASE, CARGO PARACHUTE, AIRDROP, M-2

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4531	256-4531
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-UXT Natick, MA 01760-5017	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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